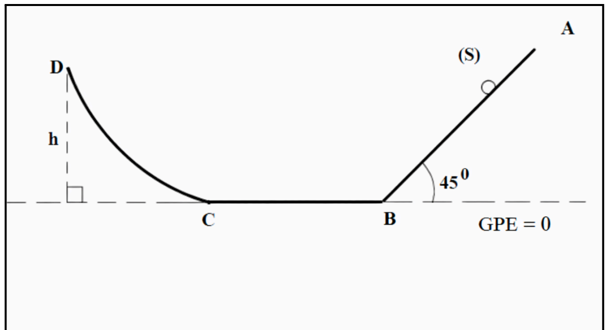
Academic Year 2025– 2026 **Physics Worksheet** **Rot. Dyn. - Work & Energy - Capacitors**		**Class: Grade 11** **Coordinateur : Racha Deguise**

Exercise 1: Work and Energy

A point mass (S), of mass m=250 g, is released without initial velocity from point A of an ABCD track located in the vertical plane. Friction is negligible on parts AB and CD of the track, but the friction on part BC is equivalent to a constant force of magnitude f = 1 N (Document 2).



Document 2

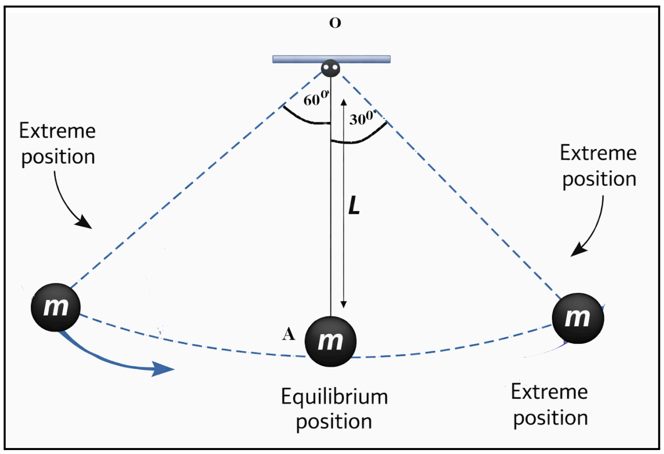
Given: α = 45°; AB = BC = 1m and g = 10m/s².

1. By applying the kinetic energy theorem to (S), calculate:
    a. The speed of (S) at B.
    b. The speed of (S) at C.
2. By using the conservation of mechanical energy, calculate the maximum height h reached by (S) at point D.
3. Show that the variation of the gravitational potential energy of (S) between C and D is equal to the opposite of the work done by the weight of (S).

We choose the horizontal plane passing through B and C as the reference level for the Gravitational Potential Energy (GPE = 0).

Exercise 2: Conservation of Mechanical Energy

A simple pendulum OA is capable of rotating around a fixed horizontal axis passing through its fixed end O. Its mass is m = 50 g and its length is L = OA = 80 cm. Starting from its vertical equilibrium position, the pendulum is deflected by an angle of 60° and then released without initial velocity at t = 0. At instant t_1, it passes through the position making an angle of 30° with the vertical. The horizontal plane passing through the axis O is chosen as the reference level for the Gravitational Potential Energy.



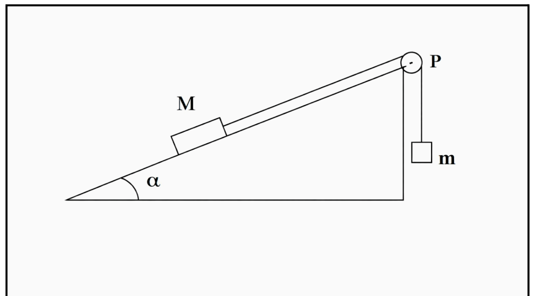
a. Calculate the ME of the system (rod; Earth) at t = 0.

b. By applying the conservation of ME, determine the velocity of the mass at instant t_1.

c. Calculate the variation of the GPE of the system (pendulum; Earth). Deduce the work done by the weight between these 2 positions.

Exercise 3: Compound System

A mass m is suspended vertically by an inextensible string of negligible mass, which is wrapped around a pulley (P) of negligible mass. The other end of the string is attached to a mass (M), which can slide along the line of greatest slope of an inclined plane making an angle α with the horizontal. (Document 4)



Document 4

We assume that friction between the support and mass M is negligible. The system (m, M, pulley) starts from rest at t = 0.

Given: m = 200 g; M = 800 g; α = 30°; g = 10 m/s^2.

1) Find, by applying the Kinetic Energy Theorem, the speed v acquired by mass m for a displacement of 10 cm.

2) Determine the value of the acceleration a of mass m, assumed to be constant. Deduce the nature of its motion.

3) Calculate the duration of mass M's displacement.

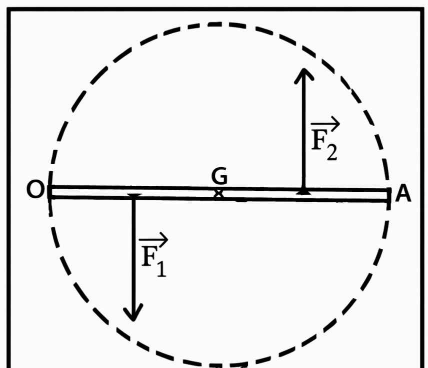
4) Calculate the power of the weight of mass M at the instant the displacement is 10 cm.

5) Calculate the average power of the weight of mass M, as well as that of the weight of mass m.

6) Determine the magnitude of the tension in each of the two segments of the string.

Exercise 4: Study of the Motion of a Rod under the Action of a Couple of Forces

The rod OA of mass M = 20 kg and length L = OA = 180 cm is capable of rotating around a vertical axis passing through its center of gravity G, under the action of a couple of forces as indicated in Document 5. Friction is negligible. The forces $\vec{F}_1$ and $\vec{F}_2$ are always horizontal, perpendicular to OA, applied respectively to the midpoints of OG and GA, and have the same constant magnitude F.



Document 5

1) By applying Newton's Second Law, specify the nature of the rod's motion.

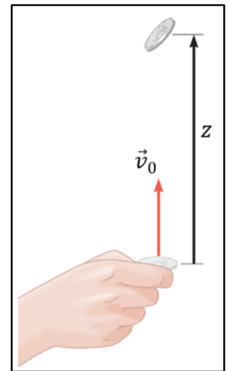
2) Starting from rest, the rod manages to perform 240 revolutions per minute over 10 s.

2-1) Deduce the angular acceleration of the rod.

2-2) Deduce the value F of the magnitude of forces $\vec{F}_1$ and $\vec{F}_2$, knowing that the moment of inertia of the rod with respect to the axis passing through G is I = 10.8 kg.m^2.

2-3) Find the rotational speed in revolutions per minute after 2 minutes from the start.

Exercise 5: Vertical Motion of an Object: Energy Analysis with and without Air Resistance



Doc. 1

An object (S) of mass $m = 0.2kg$ is launched from the ground at instant $t_0 = 0\ s$, vertically upwards with an initial velocity $\vec{v}_0$ (doc.1). Take the ground as the reference level for the GPE. Take $g = 10m.s^{-2}$.

1- Motion without air resistance

Document 2 represents the variation of the ME, GPE, and KE of the system [(S), Earth], as a function of the height z during the upward phase.

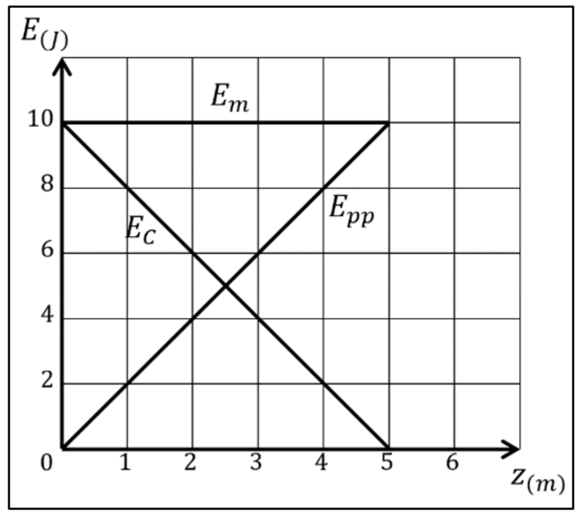
a. Show, by referring to **doc.2**, that there is no friction during the motion.

b. What is the value of the GPE at instant $t_0 = 0$.

c. Write the expression for the GPE as a function of the height z of the object (S).

d. Deduce that we can write $KE = 10 - 2z$.

e. Find graphically the value of the maximum height z_{max} reached by (S).



Doc. 2

2- Motion with air resistance

In another experiment, the object (S) is launched again. The figure of **doc.3** represents the variation of the KE and GPE of (S) during the ascent as a function of the height z.

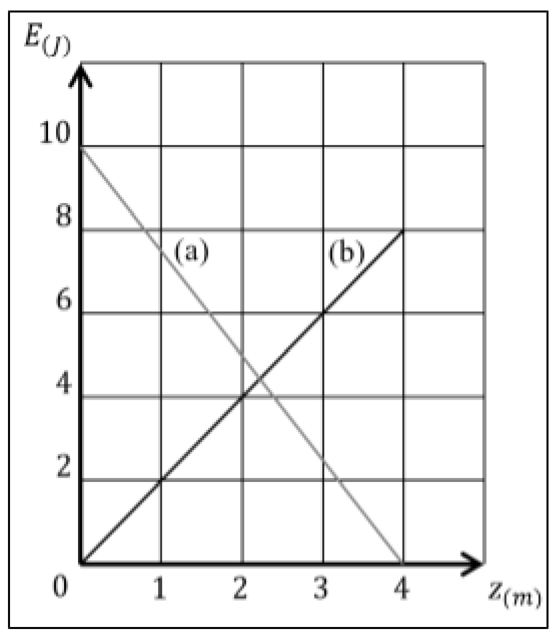
a. Curve (a) represents the variation of the KE. Justify.

b. Draw on doc.3 the curve representing the variation of the ME as a function of the height z. Deduce that the friction force exists.

c. Determine graphically the maximum height reached by (S).

d. Show that the ME decreases by $|\Delta E_m| = 2J$ between $z = 0$ and $z = z_{max}$.

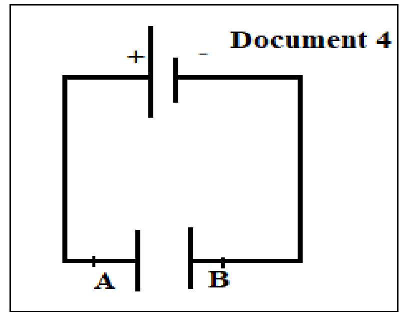
e. Calculate the value of the work done by friction during the ascent. Is this work motive or resisting? Justify.



Doc. 3

Exercise 6: Capacitors

A parallel-plate capacitor whose plates are square, with side a = 2 cm, and separated by air at a distance d = 0.2 mm (Document 4).



Given : $\varepsilon_0 = \frac{1}{36\pi.10^9} = 8.84\times10^{-12}$ (SI)

1. Calculate the capacitance C of the capacitor.
2. The capacitor of capacitance C is charged under a voltage U_{AB} = 20 V.

   2.2. Calculate the value of the charge of the capacitor.

   2.1. Deduce the charges q_A *and* q_B of each of the armatures A and B .

   2.3. Calculate the electrical energy stored by the capacitor.